WORKED SOLUTION

STEP II 2004, Question 4

Pure • Optimisation in Geometry

STEP II 2004, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

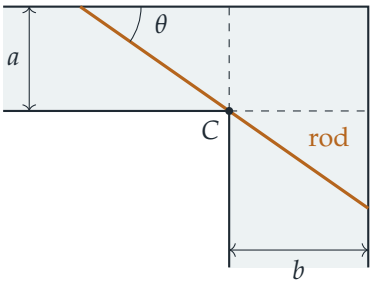
step.maths.org/questions/04-s2-q4

The key idea

Whatever happens, the rod (or table) has to swing from pointing along one corridor to pointing along the other, so at some moment it makes each angle θ in between with the walls. For a fixed angle, the most room there is lies along the line that just grazes the inner corner, and that length is easy to compute with right-angled triangles. So the object gets round only if its length is at most the *smallest* of these lengths over all angles, and calculus finds that smallest value.

Solution

Throughout, put the outer corner at the origin, with the outer walls along $y = 0$ (corridor of width a, running to the left) and $x = 0$ (corridor of width b, running downwards). The inner corner is then at $C = (-b, -a)$.



Part (i)

Suppose that at some moment the rod makes an acute angle θ with the walls of the a-corridor, sloping down from the outer wall $y = 0$ towards the outer wall $x = 0$ as in the figure.

The longest segment at angle θ. Take the line at angle θ through C. From C up to the wall

$y = 0$ is a vertical rise of a, so that piece has length $a\operatorname{cosec}\theta$; from C across to the wall $x = 0$ is a horizontal run of b, so that piece has length $b\sec\theta$. Hence the chord of this line between the two outer walls has length

$$f(\theta) = a\operatorname{cosec}\theta + b\sec\theta.$$

Any other line at angle θ that runs from one corridor into the other must pass on the outer side of C (otherwise it would cut through the inner walls). Such a line, together with the two outer walls, cuts off a smaller similar right-angled triangle, so its chord between the outer walls is shorter than $f(\theta)$. And if at this moment the rod lies wholly in the a-corridor, it fits in a strip of width a, so $L\sin\theta \le a$, giving $L \le a\operatorname{cosec}\theta < f(\theta)$; similarly $L \le b\sec\theta < f(\theta)$ if it lies wholly in the b-corridor. In every case,

$$L \le f(\theta) \quad \text{for each angle } \theta \text{ the rod passes through.}$$

As the rod turns from along the first corridor to along the second, θ varies continuously from near 0 to near $\frac{1}{2}\pi$, so it takes every value in between. Therefore $L \le f(\theta)$ for *every* $\theta \in (0, \frac{1}{2}\pi)$, that is, L is at most the minimum of f.

Minimising f. Differentiating,

$$f'(\theta) = -\frac{a\cos\theta}{\sin^2\theta} + \frac{b\sin\theta}{\cos^2\theta} = \frac{b\sin^3\theta - a\cos^3\theta}{\sin^2\theta\cos^2\theta}.$$

This is zero exactly when $\tan^3\theta = a/b$, which has a single solution α in $(0, \frac{1}{2}\pi)$, since tan is increasing there. The sign of f' is that of $b\tan^3\theta - a$, which is negative before α and positive after, so α gives the minimum. (Also $f \to \infty$ at both ends.) Taking $\theta = \alpha$ in $L \le f(\theta)$ gives

$$L \le a\operatorname{cosec}\alpha + b\sec\alpha, \qquad \tan^3\alpha = \frac{a}{b},$$

as required.

Part (ii)

Now the long sides of the table play the role of the rod. Its inner long side lies on a line at angle θ, and the outer long side lies on the parallel line a distance w further towards the outer corner. Both table corners on the outer side must lie inside the outer walls.

The room along the outer side. A line at angle θ whose distance from the outer corner O is d cuts off a right-angled triangle at O with hypotenuse c (the chord between the walls), legs $c\cos\theta$ and $c\sin\theta$, and altitude d from O. Twice its area is both $c\cos\theta \cdot c\sin\theta$ and $c\,d$, so

$$c = \frac{d}{\sin\theta\cos\theta} = 2d\operatorname{cosec}2\theta.$$

Chord length is proportional to the distance from O. Moving the line in by w (towards O) therefore shortens its chord by $2w\operatorname{cosec}2\theta$.

Ask yourself Why must the outer long side fit between the outer walls on the line that is w in from the line through C?

The inner long side cannot cross the inner corner, so it lies on the line through C or outside it. The outer long side is then on a line at least w further out, whose chord is at most $f(\theta) - 2w\operatorname{cosec} 2\theta$. (The two short sides run inwards from the outer corners, perpendicular to the long sides, and stay inside the corridors, so they impose nothing more.) If instead the whole table lies in the a-corridor, then its height across the strip is $l\sin\theta + w\cos\theta \le a$. The resulting bound $l \le (a - w\cos\theta)\operatorname{cosec}\theta$ is smaller than $g(\theta)$ below by $(b - w\sin\theta)\sec\theta > 0$, since $w < b$. The b-corridor case is the same. So, exactly as in part (i), success requires $l \le g(\theta)$ for all $\theta \in (0, \frac{1}{2}\pi)$, where

$$g(\theta) = a\operatorname{cosec}\theta + b\sec\theta - 2w\operatorname{cosec} 2\theta.$$

(The table must be narrower than both corridors, $w < a$ and $w < b$, or it cannot even travel along them.)

Minimising g. Since $\dfrac{\mathrm{d}}{\mathrm{d}\theta}\operatorname{cosec} 2\theta = -\dfrac{2\cos 2\theta}{\sin^2 2\theta}$,

$$g'(\theta) = -\frac{a\cos\theta}{\sin^2\theta} + \frac{b\sin\theta}{\cos^2\theta} + \frac{4w\cos 2\theta}{\sin^2 2\theta}.$$

Multiply by $\sin^2\theta\cos^2\theta = \frac{1}{4}\sin^2 2\theta > 0$. Then $g'(\theta) = 0$ exactly when

$$-a\cos^3\theta + b\sin^3\theta + w\cos 2\theta = 0, \qquad \text{i.e.} \qquad w\cos 2\theta = \cos^3\theta\,(a - b\tan^3\theta).$$

Write $\cos 2\theta = \cos^2\theta\,(1 - \tan^2\theta)$ and divide by $\cos^2\theta$. If $\theta = \frac{1}{4}\pi$ the equation would read $0 = \frac{1}{2\sqrt{2}}(a - b)$, which is impossible since $a \ne b$. So $\tan^2\theta \ne 1$ at a stationary point, and we may divide by $1 - \tan^2\theta$ to get

$$w = \left(\frac{a - b\tan^3\theta}{1 - \tan^2\theta}\right)\cos\theta.$$

As $\theta \to 0^+$, $g(\theta) \approx (a - w)/\theta \to +\infty$ because $w < a$; similarly $g \to +\infty$ as $\theta \to \frac{1}{2}\pi^-$ because $w < b$. So the continuous function g attains its minimum at an interior point, where $g' = 0$. Call this point β; it satisfies the displayed equation. Taking $\theta = \beta$ in $l \le g(\theta)$ gives

$$l \le a\operatorname{cosec}\beta + b\sec\beta - 2w\operatorname{cosec} 2\beta,$$

as required. (In fact $l \le \min g \le g(\beta)$ for *any* root β of the equation, so it doesn't matter which root is meant.)

Checks

- Putting $w = 0$ in part (ii) gives $\tan^3\beta = a/b$ and the bound of part (i), as it should.
- In part (i), $\sin\alpha = a^{1/3}/\sqrt{a^{2/3} + b^{2/3}}$ and $\cos\alpha = b^{1/3}/\sqrt{a^{2/3} + b^{2/3}}$, so the bound is $\left(a^{2/3} + b^{2/3}\right)^{3/2}$. For $a = 1$, $b = 8$ this is $5^{3/2} \approx 11.18$, and a numerical search for the minimum of f agrees.

- For a few hundred random $a \neq b$ and $w < \min(a, b)$, we found the minimum of g by brute force and confirmed that the minimising angle satisfies the equation for β. We also checked the chord lengths directly from coordinates.

Common slips

- Maximising f instead of minimising it. The rod has to survive the *worst* angle, so the bound is the minimum of the available length.
- Taking the table's *inner* side as the one limited by the outer walls. It is the outer side, on the line w nearer the outer corner, that is squeezed.
- Dividing by $1 - \tan^2 \beta$ without noticing that this needs $a \neq b$.

Beyond STEP

Part (i) is a classic "ladder round a corner" problem, and its answer $\left(a^{2/3} + b^{2/3}\right)^{3/2}$ is linked to the astroid: the lines through the inner corner sweep out an envelope. Part (ii) is a small case of the *moving sofa problem*: which shape of largest area can be carried round a unit-width right-angled corner? It was posed by Leo Moser in 1966 and has resisted a complete solution for most of its history. For more geometry-meets-calculus problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.